Django REST Framework (DRF) Roadmap
Section 1: Introduction

Main Topics

• What is an API?

• What is REST?

• What is Django REST Framework?

• Why use DRF instead of plain Django?

• Client-Server Architecture

• Request-Response Cycle

• HTTP Protocol Basics

• HTTP Methods

• HTTP Status Codes

• JSON Fundamentals

• API Endpoints

• RESTful API Principles

• Understanding CRUD Operations

• Installing DRF

• Setting up your First DRF Project

Subtopics

• API Definition, Real-world Examples, Frontend ↔ Backend Communication

• REST Meaning, Stateless Communication, Resource-Based Architecture

• Client, Server, Database, API Layer

• Request, Processing, Response, Response Formats

• HTTP Methods: GET, POST, PUT, PATCH, DELETE

• Status Codes: 200, 201, 204, 400, 401, 403, 404, 500

• JSON Objects, Arrays, Key-Value Pairs, Nested JSON

• API Endpoints: /api/products/, /api/categories/, /api/users/

• CRUD Mapping: Create(POST), Read(GET), Update(PUT/PATCH), Delete(DELETE)

• Installing and Configuring DRF

Practical Exercise

• Create a Django project named ecommerce.

• Create a products app.

• Install djangorestframework.

• Add rest_framework to INSTALLED_APPS.

• Run the development server.

eCommerce Project Application

• Understand APIs

• Understand frontend-backend communication

• Understand REST principles

• Understand JSON structure

• Understand HTTP methods and status codes

• Successfully install and configure DRF

Mini Project Task

• Create the ecommerce project structure.

• Do not create models yet.

• Do not create serializers yet.

• Focus on understanding the DRF ecosystem.

Next Section

• Django Project Structure & DRF Setup

• Project vs App

• settings.py

• urls.py

• manage.py

• DRF Configuration

• First Endpoint

• First API Response